MOVIE TICKET BOOKING

Report

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR MAJOR PROJECT

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ABSTRACT

The Movie Booking System is a user-friendly software application developed to streamline the process of selecting and booking movie tickets. Designed primarily for cinema operators and customers, the system allows users to browse available movies, check showtimes, select preferred seats, and complete bookings efficiently.

The system replaces traditional manual ticketing processes with a digital solution, reducing human error and saving time for both staff and customers. It offers functionalities such as movie selection, real-time seat availability, customer registration/login, booking history, and ticket confirmation. Administrators can update movie listings, manage show timings, and monitor bookings through a secure backend interface.

This project is developed using Python and Tkinter for the front-end GUI, with data stored using file handling or database integration (like SQLite/MySQL). It emphasizes user interaction, data validation, and smooth navigation to ensure a seamless user experience.

The Movie Booking System not only demonstrates core programming concepts like OOP, file/database operations, and GUI development but also simulates a real-world application that highlights the benefits of automation in the entertainment industry.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me in the successful completion of this project titled "Movie Booking System".

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solving skills.

TABLE OF CONTENTS

Contents: Page no.

1. Abstract
2. Acknowledgement
3. Table of Contents
4. Company Profile

Chapter 1: Introduction
1.1Introduction to Project

1.2 Problem Definition

1.3 Objectives

Chapter 2:Implementation,Testing and Maintenance
2.1Introduction to Languages

2.2Introduction to IDEs

CHAPTER 3:Requirement Analysis and System Specifications
3.1 Feasibility study(Technical,Economical,Operational)

3.2 SDLC Model

Chapter 4:Hardware and Software Requirements

4.1 Hardware Requirements

4.2 Software Requirements

4.3 Tools used for development of Project

4.4 DFDS

4.5 Flow Chart

CHAPTER 5: Implementation,Testing and Maintenance

5.1 Implementations

5.2 Testing

5.3 Maintenance

CHAPTER 6: RESULTS

Snapshots

CHAPTER 7:Conclusions and Future Scope

7.1 Summary

7.2 Conclusion

7.3 Future Scope

CHAPTER 8:Bibliography and Conclusion

INTRODUCTION TO THE INSTITUTION CONTENTS

O7 Services is an ISO 9001:2015 certified company, established in 2015, with a strong reputation for delivering end-to-end IT solutions. The company specializes in a wide range of services, including Web Development, Mobile Application Development, Custom Software Development, UI/UX Design, Hosting Services, Digital Marketing, Domain Registration across multiple extensions, AMC & MMC services, Bulk SMS, and Voice Call solutions.

With 10+ years of experience, O7 Services provides complete business solutions—from consulting and development to deployment, QA, and 24/7 support. Known for timely delivery and affordable pricing, the company builds strong, long-term partnerships. It is based in Jalandhar with a branch in Hoshiarpur.

O7 Services has developed a wide range of software products, including Vehicle Tracking Systems, Invoice Software, School and Hospital Management Systems, Parent-Teacher Communication Apps, Fee and Task Management Systems, Online Food Ordering Applications, Security Apps, Admission Systems, Inventory Software, and Car Servicing Apps. Each solution is designed to be practical, reliable, and user-friendly, addressing real-world business needs effectively.

O7 Services also offers industry-focused training programs, including 6-week and 6-month industrial training, project-based training, corporate training, and job-oriented courses, all tailored to meet current industry needs. The training covers a broad spectrum of modern technologies, such as Full Stack Development (MEAN/MERN), Flutter, Kotlin Android, Swift UI (iOS), Firebase, Python, Angular, React.js, Vue.js, Node.js, ASP.NET, .NET Core, PHP, Laravel, CodeIgniter, Software Testing, Cloud Computing, Blockchain, DevOps, Data Science, Graphic Designing, Artificial Intelligence, Machine Learning, UI/UX Design, Digital Marketing,

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CHAPTER 1:INTRODUCTION

1.1:INTRODUCTION

In the digital era, the demand for online services has increased significantly, and the movie industry is no exception. A Movie Ticket Booking System is a software-based solution that allows users to view current movies, check showtimes, select seats, and book tickets from the comfort of their homes or via a computer interface.

Traditionally, booking movie tickets involved visiting the cinema hall and standing in long queues, which was time-consuming and inconvenient. With the introduction of automated booking systems, this process has become much faster, more efficient, and accessible to everyone.

This project aims to develop a user-friendly application where users can:

- Register and log in securely
- Browse movies and show timings
- Check seat availability
- Select desired seats
- Confirm and view booking details

The system also supports an admin interface where cinema staff can:
Add or remove movies

- Set showtimes
- View overall bookings

The Movie Ticket Booking System not only simplifies the movie-watching experience but also demonstrates practical implementation of programming concepts such as GUI design, file/database handling, user input validation, and system logic.

This project is developed using Python with Tkinter for the graphical user interface, and it is ideal for learning how real-world applications are built and managed.

In today's fast-paced digital world, the traditional method of booking movie tickets by standing in long queues at cinema halls has become outdated and inefficient. With the rise of technology, especially in the field of software development and internet services, there is an increasing demand for convenient, quick, and hassle-free ticket booking systems. The Movie Ticket Booking System is developed to meet this demand by offering a computerized and automated solution for booking movie tickets.

The system provides a platform where users can browse a list of currently running movies, view their showtimes, and book tickets according to their convenience. It eliminates the need for physical presence, reduces manual errors, improves seat management, and enhances the overall movie-going experience. Users can log into the system, check the availability of seats for a particular show, choose their desired seats, and make a booking within minutes.

From a technical perspective, the system is built using Python programming language and uses the Tkinter library for designing a graphical user interface (GUI). For data management, it can utilize file handling or connect to a database (e.g., SQLite or MySQL) to store user details, movie lists, bookings, and showtimes. The application follows a modular approach where features like login/registration, movie listing, seat selection, and ticket confirmation are separated for better maintainability and clarity.

1.2:Problem Definition

Problems Addressed by the System

- Manual Booking Hassles

Traditional ticket booking requires users to visit the cinema, stand in long queues, and face delays, especially during weekends or holidays.

- Seat Availability Confusion

Without a digital system, customers often cannot view real-time seat availability, leading to miscommunication or overbooking.

- Inefficient Record Keeping

Manual record-keeping of bookings is prone to errors, loss of data, and difficulty in tracking customer history or show performance.

- Limited Access & Timing

Manual booking counters operate during fixed hours. Users cannot book tickets at their convenience (e.g., at night or remotely).

- Lack of User Convenience

Users often have limited information about available movies, showtimes, or seat layout when booking manually.

- No Personalization

Manual systems don't store user data, so customers need to repeat their information every time they book.

The traditional method of booking movie tickets is outdated and inefficient in today's fast-paced digital world. Customers are often required to visit cinema halls physically, stand in long queues, and wait for long durations to purchase tickets. This manual approach not only consumes time but is also prone to human errors, mismanagement of seat allocations, and lack of real-time

updates. The absence of an automated system results in customer dissatisfaction and operational inefficiencies for cinema management.

Furthermore, users lack access to important features such as seat availability visualization, flexible showtime selection, and instant booking confirmation. Administrators also struggle with maintaining records, tracking user data, and updating movie schedules manually.

Thus, there is a strong need for a Movie Ticket Booking System that can automate the entire ticketing process, provide a user-friendly interface, ensure accurate seat management, and simplify administration tasks.

1.3 Objectives

The primary objective of a movie ticket booking system project in Python is to create a user-friendly platform for booking movie tickets online, offering features like movie browsing, seat selection, online payment, and booking management. This system aims to streamline the booking process, reduce manual effort for theater staff, and provide convenient anytime-anywhere access for customers.

Core Objectives:

- Efficient Online Booking: Enable users to browse movie listings, view showtimes, select seats, and purchase tickets online.
- User-Friendly Interface: Design a simple and intuitive interface for both customers and administrators.
- Real-time Seat Availability: Provide up-to-date information on seat availability to avoid overbooking.
- Secure Online Payment: Integrate secure payment gateways for online transactions.
- Booking Management: Allow users to view, manage, and cancel their bookings.
- Admin Panel: Provide an administrative interface for managing movies, showtimes, ticket pricing, and user data.
- Reduce Manual Work: Minimize the need for manual ticketing operations at the theater.
- Enhanced Customer Experience: Offer a convenient and efficient way for customers to book tickets.

CHAPTER 2

2.1 Introduction to Languages:

Python is a high-level, interpreted programming language known for its simplicity and readability. Python emphasizes code readability and clarity, making it an excellent choice for beginners and experienced programmers alike.

- Web Applications: Python is widely used for developing web applications due to its simplicity and the availability of powerful frameworks like Django, Flask, and Pyramid. These frameworks provide tools and libraries for building scalable and feature-rich web applications, including content management systems (CMS), e-commerce platforms, social media platforms, and more. Python's integration with web technologies such as HTML, CSS, and JavaScript makes it a popular choice for full- stack web development.

- Desktop GUI Applications: Python can be used to develop desktop GUI applications using libraries such as Tkinter, PyQt, and wxPython. These libraries provide a set of tools and widgets for creating interactive graphical user interfaces (GUIs) for applications ranging from simple utilities to complex desktop software. Python's simplicity and cross- platform compatibility make it an excellent choice for developing desktop applications that can run on Windows, macOS, and Linux operating systems.

- Scripting and Automation: Python's scripting capabilities make it well-suited for automating tasks and writing scripts for system administration, software testing, and workflow automation. Python's standard library includes modules for interacting with the file system, network sockets, databases, and other system resources, making it a powerful tool for scripting tasks. Python's ease of use and readability make it an ideal choice for writing scripts that automate repetitive tasks and streamline workflows in various domains

Version:-

Table 2.1: - Versions of Python

Name	**Python Version**	**Set up tools version**	**Pip Version**	**Results**
System Python	2.7.12	42.0.2	8.1.1	PASS
System Python 3	3.5	42.0.2	8.1.1	PASS
Virtualenv Python 3	3.5	20.7.0	19.3.1	PASS
Virtualenv Python 3	2.7.12	42.0.2	19.3.1	PASS
Virtualenv Python 2	2.7.12	42.0.2	8.1.1	PASS
Virtualenv Python 2	2.7.12	42.0.2	19.3.1	FAIL
Virtualenv Python 2	2.7.12	20.7.0	19.3.1	FAIL

Characteristics of Python:

1. Simple and Easy to Learn**:** Python has a clean and straightforward syntax, making it easy for programmers to grasp concepts quickly. Its readability reduces the cost of program maintenance and development

2. Interpreted Language: Python is an interpreted language, meaning that code written in Python is executed line by line, making the development process more agile and allowing for easier debugging.

3. High-level Language: Python abstracts low-level details, allowing programmers to focus on solving problems rather than worrying about system-level management.

4. Dynamic Typing**:** Python uses dynamic typing, where the type of a variable is determined at runtime. This makes Python flexible and versatile, but it also requires careful attention to variable types in large projects.

5. Extensive Standard Library: Python comes with a comprehensive standard library that provides support for a wide range of tasks, from file I/O to networking to data manipulation. This reduces the need for external libraries for many common programming tasks.

6. Multi-Programming**:** Python supports multiple programming, including procedural, object-oriented, and functional programming. This versatility allows developers to choose the best approach for solving a particular problem.

7. Platform Independent: Python code can run on various operating systems, including Windows, macOS, and Linux, without modification. This platform

independence enhances its portability and makes it suitable for cross-platform development.

8. Large Community and Ecosystem: Python has a vast and active community of developers who contribute to its growth and development. This community support is evident in the extensive range of third-party libraries and frameworks available for various applications, such as web development, data analysis, machine learning, and more.

9. Open Source: Python is open-source software, meaning its source code is freely available for anyone to use, modify, and distribute. This fosters collaboration and innovation within the Python community.

2.1.1 Features of Python:

- Embeddable
  - Python can be used within C/C++ program to give scripting capabilities for the program's users.
- Robust:
  - Exceptional handling features
  - Memory management techniques in built

2.1.2 Advantages of Python:

1. Readability**:** Python's clean and simple syntax emphasizes readability, making it easier for developers to write and maintain code. This readability reduces the time and effort required for understanding and debugging code, particularly in large projects or when collaborating with other developers.

2. Versatility: Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming, allowing developers to choose the best approach for solving a particular problem. This versatility makes Python suitable for a wide range of applications, from web development to scientific computing to artificial intelligence.

2.2 INTRODUCTION TO IDE

An Integrated Development Environment (IDE) is a software application that provides comprehensive facilities to computer programmers for software development. It typically includes a source code editor, build automation tools, and a debugger, along with other features that streamline the development process. IDEs are designed to enhance productivity by providing an all-in-one environment for writing, testing, and debugging code.

Some popular IDEs for Python development include:

1. PyCharm: PyCharm is a powerful Integrated Development Environment (IDE) developed by JetBrains specifically for Python development. It offers a comprehensive set of features designed to enhance productivity and streamline the development process for Python programmers.

2. Spyder: Spyder is an open-source Integrated Development Environment (IDE) specifically designed for scientific computing and data analysis with Python. Developed by the Spyder Project, it provides a powerful set of tools tailored to the needs of scientists, engineers, and data analysts.

3. Visual Studio Code: Visual Studio Code (VS Code) is a lightweight, open-source code editor developed by Microsoft. While it's not a full-fledged IDE like PyCharm, it offers many features that make it a powerful tool for Python development.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Feasibility Study-

- Executive Summary:
- Briefly summarize the key aspects of the feasibility study, highlighting the potential benefits of implementing a Traffic Analysis AI System for urban transportation.

❖ Market Feasibility:

- Analyze the demand for intelligent traffic analysis solutions in urban environments.
- Identify potential stakeholders, including government agencies, urban planners, and transportation authorities.
- Assess market trends, competition, and potential user acceptance.

❖ Technical Feasibility:

- Evaluate the technical requirements for implementing the AI system.
- Assess the availability of necessary technology, infrastructure, and expertise.
- Identify potential challenges related to data processing, computer vision, and system integration.

❖ Financial Feasibility:

- Estimate the initial and ongoing costs associated with developing, implementing, and maintaining the Traffic Analysis AI System.
- Conduct a cost-benefit analysis to determine the financial viability.
- Project potential revenue streams and return on investment over time.

❖ Operational Feasibility:

- Assess how well the proposed AI system integrates with existing traffic management systems and processes.
- Evaluate the impact on day-to-day operations for traffic authorities and other stakeholders.
- Identify potential operational challenges and propose solutions.

❖ Legal and Regulatory Feasibility:

- Examine legal and regulatory requirements related to traffic data collection, processing, and privacy.
- Identify any permits, licenses, or compliance issues that need to be addressed.
- Ensure that the system aligns with data protection regulations.

❖ Scheduling Feasibility:

- Develop a realistic project timeline with key milestones and deadlines.
- Identify critical tasks and potential bottlenecks.
- Evaluate the feasibility of meeting project deadlines.

❖ Risk Analysis:

- Identify and assess potential risks associated with the development and implementation of the Traffic Analysis AI System.
- Develop strategies for mitigating and managing identified risks.

- Consider risks related to data security, system reliability, and algorithm accuracy.

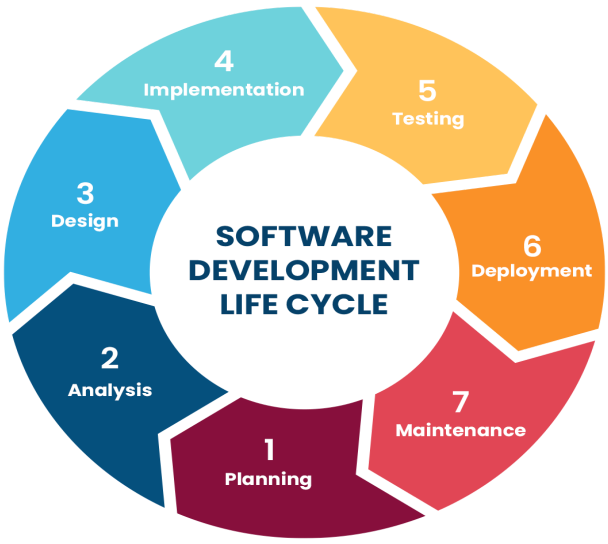
❖ Recommendations:

- Summarize the findings of the feasibility study.
- Provide clear recommendations on whether to proceed with the development and implementation of the Traffic Analysis AI System

3.2 SDLC Model

SDLC is a process followed for software building within a software organization. SDLC consists of a precise plan that describes how to develop, maintain, replace, and enhance specific software. The life cycle defines a method for improving the quality of software and the all- around development process.

SDLC specifies the task(s) to be performed at various stages by a software engineer or developer. It ensures that the end product is able to meet the customer's expectations and fits within the overall budget. Hence, it's vital for a software developer to have prior knowledge .



Stage-1: Planning and Requirement Analysis

Planning is a crucial step in everything, just as in software development.In this same stage, requirement analysis is also performed by the developers of the organization. This is attained from customer inputs, and sales department/market surveys.The information from this analysis forms the building blocks of a basic project. The quality of the project is a result of planning. Thus, in this stage, the basic project is designed with all the available information.

Stage-2: Defining Requirements

In this stage, all the requirements for the target software are specified. These requirements get approval from customers, market analysts and stakeholders. This is fulfilled by utilizing SRS (Software Requirement Specification). This is a sort of document that specifies all those things that need to be defined and created during the entire project cycle.

Stage-3: Designing Architecture

SRS is a reference for software designers to come up with the best architecture for the software. Hence, with the requirements defined in SRS, multiple designs for the product architecture are present in the Design Document Specification (DDS).This DDS is assessed by market analysts and stakeholders. After evaluating all the possible factors, the most practical and logical design is chosen for development.

Stage-4: Developing Product

At this stage, the fundamental development of the product starts. For this, developers use a specific programming code as per the design in the DDS. Hence, it is important for the coders to follow the protocols set by the association. Conventional programming tools like compilers, interpreters, debuggers, etc. are also put into use at this stage. Some popular languages like C/C++, Python, Java, etc. are put into use as per the software regulations.

Stage-5: Product Testing and Integration

After the development of the product, testing of the software is necessary to ensure its smooth execution. Although, minimal testing is conducted at every stage of SDLC. Therefore, at this stage, all the probable flaws are tracked, fixed, and retested. This ensures that the product confronts the quality requirements of SRS.

Documentation, Training, and Support: Software development is an essential part of the software development life cycle. A well-written document acts as a tool and means to information repository necessary to know about software

processes, functions, and maintenance. Documentation also provides information about how to use the product. Training in an attempt to improve the current or future employee performance by increasing an employee's ability to work through learning, usually by changing his attitude and developing his skills and understanding.

Stage 6: Deployment and Maintenance of Products

After detailed testing, the conclusive product is released in phases as per the organization's strategy. Then it is tested in a real industrial environment. It is important to ensure its smooth performance. If it performs well, the organization sends out the product as a whole. After retrieving beneficial feedback, the company releases it as it is or with auxiliary improvements to make it further helpful for the customers. However, this alone is not enough. Therefore, along with the deployment, the product supervisors.

CHAPTER 4:

HARDWARE AND SOFTWARE REQUIREMENTS

- HARDWARE REQUIREMENTS: -

Hardware Minimum Hardware Requirements

- Processor i5 processor system or higher
- RAM 8 Image Creator from Microsoft Designer (bing.com)GB or higher
- ROM 100 GB or higher
- GeForce RTX graphics cards
- SSD 128 GB
- Windows Windows7 or higher

- SOFTWARE REQUIREMENTS: -

○	Browser-	Mozilla Firefox/Chrome/Edge
○	Operating System-	Windows Operating System/Linux
○	Text Editor-	Visual Studio Code

- TOOLS USED FOR DEVELOPMENT OF PROJECT

Python: Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and

packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributedOften, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault. Instead, when the interpreter discovers 14 an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace. A source level debugger allows inspection of local and global variables, evaluation of arbitrary expressions, setting breakpoints, stepping through the code a line at a time, and so on. The debugger is written in Python itself, testifying to Python's introspective power. On the other hand, often the quickest way to debug a program is to add a few print statements to the source: the fast edit-testdebug cycle makes this simple approach very effective.

Pandas: Pandas is a python package providing fast, flexible, and expressive data structures designed to make working with “relational” or “labelled” data both easy and intuitive. It aims to be the fundamental high-level building block for doing practical, real-world data analysis in Python. Additionally, it has the broader goal of becoming the most powerful and flexible open- source data analysis/manipulation tool available in any language. It is already well on its way toward this goal.The two primary data structures of pandas, Series (1-dimensional) and DataFrame (2-dimensional) handle the vast majority of typical use cases in finance, statistics, social science, and many areas of engineering. Here are just a few of the things that pandas does well:

- Easy handling of missing data (represented as NaN) in floating point as well as non-floating point data
- Size mutability: columns can be inserted and deleted from DataFrame and higher dimensional objects
- Automatic and explicit data alignment: objects can be explicitly aligned to a set of labels, or the user can simply ignore the labels and let Series, DataFrame, etc. automatically align the data for you in computations

● Powerful, flexible group by functionality to perform split-apply-combine operations on data sets, for both aggregating and transforming data

● Make it easy to convert ragged, differently-indexed data in other Python and NumPy data structures into Data Frame objects

● Intelligent label-based slicing, fancy indexing, and subsetting of large data sets

● Intuitive merging and joining data sets

● Flexible reshaping and pivoting of data sets

● Hierarchical labeling of axes (possible to have multiple labels per tick)

● Robust IO tools for loading data from flat files (CSV and delimited), Excel files, databases, and saving / loading data from the ultrafast HDF5 format.

● Time series-specific functionality: date range generation and frequency conversion, moving window statistics, date shifting, and lagging.

NumPy: NumPy is a Python library for numerical computing that provides support for large, multidimensional arrays and matrices, along with a collection of high-level mathematical functions to operate on these arrays. It is a fundamental library in the scientific computing ecosystem in Python, and is widely used in fields such as data science, machine learning, physics, and engineering. Some of the key features of NumPy include:

● A powerful N-dimensional array object, which allows fast and efficient manipulation of arrays and matrices

● A rich set of functions for performing mathematical operations on arrays, such as element-wise operations, matrix multiplication, and linear algebra operations

● Efficient handling of large data sets and complex calculations using optimized C code and memory management techniques

● Easy integration with other Python libraries, such as Pandas, Matplotlib, and scikit-learn.

● NumPy is also highly extensible, allowing developers to build their own functions and extensions using Python or other programming languages such as C or Fortran.

Streamlit: Streamlit is an open-source Python library that allows you to create interactive web applications for data visualization, machine learning, and data science. It provides a simple and intuitive way to build user interfaces without requiring extensive knowledge of web development technologies like HTML, CSS, or JavaScript. Here are some key features and benefits of Streamlit:

● Easy to Learn and Use: Streamlit has a very simple and straightforward syntax, making it easy for developers, especially those with a Python background, to quickly build interactive applications.

● Rapid Prototyping: With Streamlit, you can quickly prototype and iterate on your applications, as it allows you to see changes in real-time without having to reload the application.

● Interactive Data Visualization: Streamlit integrates well with popular data visualization libraries like Matplotlib, Plotly, and Altair, allowing you to create interactive and visually appealing plots and charts.

● Machine Learning Model Deployment: Streamlit provides a convenient way to deploy and showcase machine learning models by creating interactive user interfaces around them.

● Caching: Streamlit supports caching, which can significantly improve the performance of your applications by avoiding redundant computations

● File Upload and Download: Streamlit allows users to upload and download files directly from the application, making it easy to work with data files. Streamlit has gained popularity in the data science and machine learning communities due to its simplicity and ease of use. It allows researchers, analysts, and developers to quickly create interactive applications for data exploration, model visualization, and deployment, without the need for extensive web development skill.

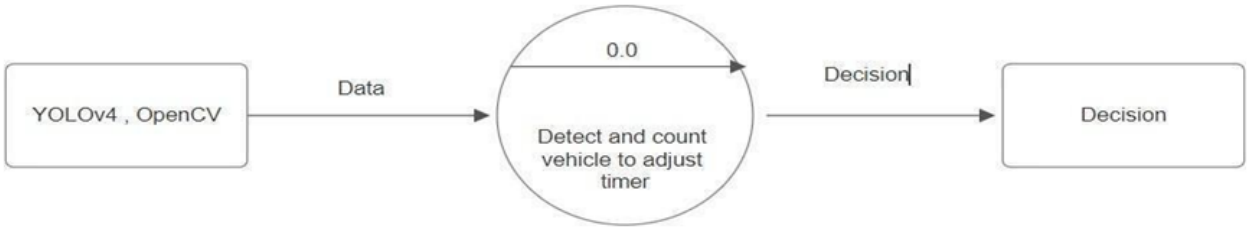
Plotly: Plotly can be extremely helpful in visualizing and analyzing the impact by turning complex data into clear, interactive visuals. It can create line charts to show how cases, deaths, and recoveries change daily or weekly. Use bar charts or scatter plots to compare data between countries or states. Plotly allows you to create global heatmaps to show the

spread of the virus. Track how many people are vaccinated in each country, and compare with case or death rates.Visualize ICU usage and bed availability or ventilator needs using time-series or bar charts.

Matplotlib: Matplotlib is one of the most widely used in Python for making charts and graphs. It's like the foundation for many other visualization tools including Seaborn. With Matplotlib can create a wide range of static (non-interactive) plots such as line graphs, bar charts, pie charts, histograms and scatter plots. It’s very flexible and powerful, but it can be a little more complex compared to Seaborn or Plotly, especially for beginners. Matplotlib is great when you need full control over your chart. You can customize almost everything—colors, labels, legends, font sizes, figure size and more. While Matplotlib is powerful, its default visuals are quite basic, which is why people often use Seaborn or Plotly for better-looking or interactive charts. Still, Matplotlib is very useful when building scientific plots, technical charts, or fine-tuned custom visuals.

4.4 DFD

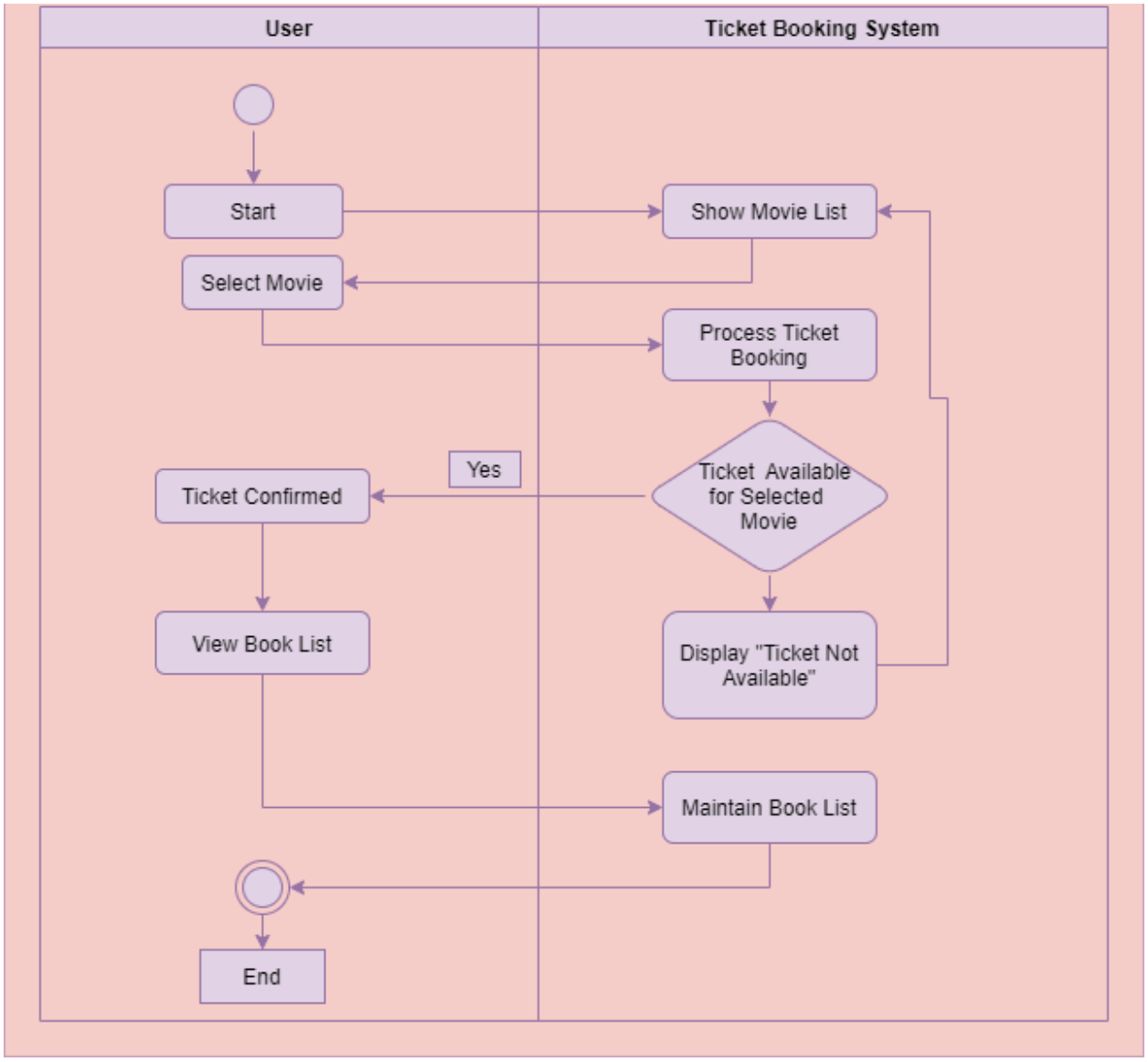
Data Flow Diagram (DFD) represents the flow of data within information systems. Data Flow Diagrams (DFD) provide a graphical representation of the data flow of a system that can be understood by both technical and non-technical users. The models enable software engineers, customers, and users to work together effectively during the analysis and specification of requirements.



4.5 Flow Chart-

Flowcharts are nothing but the graphical representation of the data or the algorithm for a better understanding of the code visually. It displays step-by-step solutions to a problem, algorithm, or process. It is a pictorial way of representing steps that are preferred by most beginner-level programmers to understand algorithms of computer science, thus it contributes to troubleshooting the issues in the algorithm. A flowchart is a picture of boxes that indicates the

process flow sequentially. Since a flowchart is a pictorial representation of a process or algorithm, it's easy to interpret and understand the process.



Implementation

The implementation of the Movie Ticket Booking System involves the development of a graphical user interface (GUI), along with backend logic to manage users, movie listings, seat availability, and bookings. The project is built using Python with the Tkinter library for the GUI and uses either file handling or SQLite for data storage.

The implementation is divided into several functional modules, each performing a specific task to ensure smooth and organized operation of the system.

Testing

Testing is a crucial phase in the software development lifecycle that ensures the system performs as expected and handles both valid and invalid inputs gracefully. The **Movie Ticket Booking System** was tested through various manual and functional testing methods to identify and fix bugs, validate inputs, and verify feature accuracy.

Types of Testing Performed

1. **Unit Testing**

○ Individual modules like login, registration, seat booking, and movie selection were tested separately to ensure each component worked correctly.

○ Example: Checked that the register_user() function does not allow duplicate usernames.

2. **Functional Testing**

○ Verified that all the core features perform as intended.

○ Scenarios tested:

- Booking a seat updates the system correctly.
- Registration form checks for empty fields.
- Login with correct/incorrect credentials.
- Selected movie and seat are correctly passed to confirmation.

3. **Boundary Testing**

○ Tested the limits of input fields such as long usernames, empty seat selections, and invalid seat names.

○ Example: Entering a seat not listed in the options results in no booking.

4. **Integration Testing**

○ Tested how modules interact — for example, after login, does it correctly pass to movie selection? Does the selected movie persist through seat selection?

5. **Error Handling Testing**

○ Verified proper error messages for:

- ■ Wrong username/password
- ■ No seat selected
- ■ Duplicate registration

6. **GUI Usability Testing**

○ Ensured that:

- ■ Buttons work and are labeled clearly.
- ■ Windows resize properly (if allowed).
- ■ Font and layout are readable and intuitive.
- ■ Navigation between windows is smooth.

Test Techniques

Testing techniques are methods used to identify defects and ensure that the software works as expected. For the **Movie Ticket Booking System**, a variety of testing techniques were used to thoroughly evaluate the system's functionality, performance, and reliability.

1. Black Box Testing

- Definition: Focuses on testing the system's functionality without knowing the internal code.
- Application in Project:

○ Tested login functionality with correct and incorrect credentials.

○ Verified movie selection and seat booking without needing to see the logic behind them.

○ Checked whether confirmation messages appeared for correct inputs.

2. White Box Testing (Code-Based Testing)

- Definition: Tests the internal logic, paths, and structure of the code.
- Application in Project:

○ Checked all if-else conditions for user authentication.

○ Verified function-level operations (e.g., what happens if a user already exists).

○ Used print/debug statements to trace variable values.

3. Positive Testing

● Definition: Testing with valid inputs to ensure the system behaves correctly.

● Application in Project:

○ Entered correct usernames and passwords.

○ Selected available seats and confirmed bookings.

○ Verified successful registration of new users.

4. Negative Testing

● Definition: Testing with invalid inputs to ensure the system handles errors properly.

● Application in Project:

○ Tried logging in with wrong passwords.

○ Left input fields blank during registration.

○ Clicked "Confirm Booking" without selecting a seat.

5. Boundary Value Testing

●Definition: Testing the system at the edge of input ranges.

● Application in Project:

○ Entered usernames with minimum and maximum characters.

○ Tested if the system restricts booking unavailable or invalid seat IDs (e.g., "Z9").

6. Integration Testing

● Definition: Testing how different modules work together.

● Application in Project:

○ Checked flow from login → movie selection → seat booking → confirmation.

○ Ensured data (like movie name and seat number) passed correctly across screens.

7. Usability Testing

- Definition: Tests user-friendliness and clarity of the interface.
- Application in Project:
  - ○ Ensured GUI elements were readable and clickable.
  - ○ Verified that navigation was smooth and logical.
  - ○ Tested the system with different users (friends/teachers) for feedback.

Test Planning

Test Planning is the process of defining the scope, objectives, approach, resources, and schedule of testing activities. It provides a roadmap for testing and ensures that all components of the system are verified thoroughly.

The Movie Ticket Booking System test plan outlines the strategy and methodology used to ensure that the software meets its intended requirements, performs reliably, and provides a good user experience.

Test Objectives

- To verify that all core functionalities (registration, login, movie selection, seat booking, and confirmation) work correctly.

- To validate that the system handles invalid inputs and user errors gracefully.

- To ensure data is stored and retrieved accurately from files or the database.

- To confirm the GUI is user-friendly, navigable, and responsive.

Test Design

Test Design is the process of identifying test conditions, creating test cases, and defining the data and steps required to validate system functionality. It ensures that each feature of the system is tested against defined requirements, both under normal and abnormal conditions.

The Movie Ticket Booking System is tested using both positive and negative test cases, and the design focuses on modular components such as registration, login, movie selection, seat booking, and booking confirmation.

CHAPTER 6:SCREENSHOTS

CHAPTER - 7
SUMMARY AND CONCLUSIONS

7.1 Summary

The Movie Ticket Booking System is a Python-based desktop application developed using the **Tkinter library**. The main objective of this project is to automate the traditional movie ticket booking process by providing a user-friendly interface where users can register, log in, view currently playing movies, select showtimes, choose available seats, and confirm bookings.

Throughout the development of this system, various programming concepts such as **object-oriented programming, GUI design, file/database handling, user input validation,** and **modular programming** were implemented. The system is designed to simplify the booking process for customers while providing basic administrative functions (optional) for managing movie and show details.

Testing techniques such as **black box testing, white box testing, boundary value analysis,** and **manual functional testing** were used to ensure the application works correctly and handles errors properly. All core modules—user registration, login, movie selection, seat booking, and confirmation—have been thoroughly tested and validated.

This project not only enhances technical skills in Python programming and GUI development but also simulates how real-world systems operate in the entertainment industry. It serves as a strong foundation for implementing advanced features in the future, such as online payment integration, real-time seat maps, or multi-screen support.

7.2 CONCLUSIONS

The Movie Ticket Booking System successfully fulfills its objective of providing a simple, efficient, and user-friendly platform for booking movie tickets. By automating the traditional

manual booking process, this system enhances user convenience, minimizes human error, and streamlines the management of movie showtimes and seat allocations.

Developed using Python and Tkinter, the system allows users to register, log in, browse movie listings, select seats, and confirm bookings—all through an interactive graphical interface. The project also demonstrates how core programming concepts like input validation, file/database handling, modular design, and GUI development can be applied to solve real-world problems.

During the course of this project, thorough testing was conducted to ensure the system performs reliably under different input conditions. It handles both valid and invalid data gracefully, offering proper error messages and feedback to users.

This project not only strengthens practical coding skills but also gives valuable insight into how real-life applications are planned, developed, and tested. It can be further improved in the future by integrating features like online payments, SMS/email confirmation, or cloud-based seat synchronization for real-time updates across multiple systems.

In conclusion, the Movie Ticket Booking System is a solid demonstration of how software development can improve efficiency, user experience, and system reliability in the cinema and entertainment domain.

These future enhancements can greatly increase the usability, efficiency, and commercial viability of the Movie Ticket Booking System. With additional features and cloud integration, the system can evolve into a professional-level platform used by cinemas and multiplexes worldwide.

7.3 Future Scope

While the current version of the **Movie Ticket Booking System** successfully meets its basic objectives, there are several enhancements and advanced features that can be implemented to

improve functionality, usability, and scalability in the future. These improvements can transform the system from a basic project into a fully functional commercial-grade application.